

CO3 – AT2 CIRCUIT DEBUGGING

Question

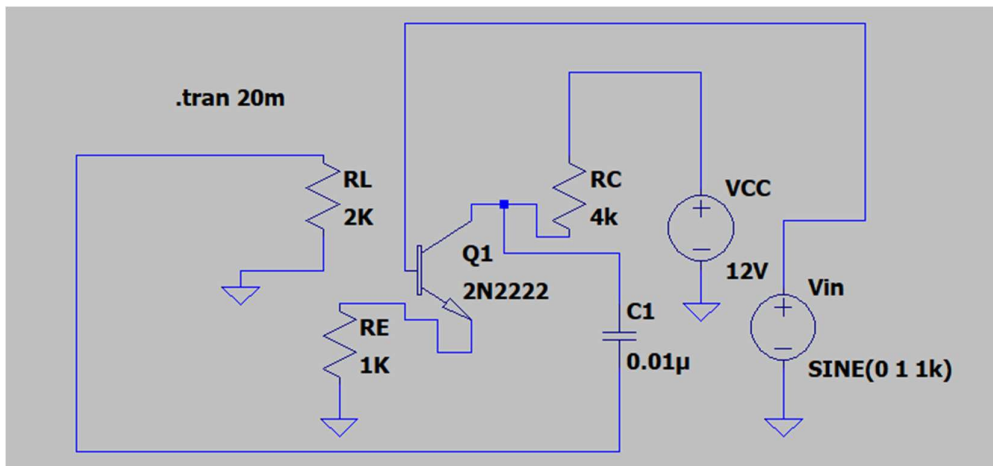
The output coupling capacitor ($C2 = 10\ \mu\text{F}$) is accidentally replaced with $0.01\ \mu\text{F}$ during simulation. Perform a transient analysis in LTspice and observe the output waveform across the load resistor. Explain how the reduced capacitor value affects the low-frequency response and the amplitude of the output signal. Discuss the role of the coupling capacitor in blocking DC while allowing AC signals to pass and justify the correct capacitor selection.

Problem statement

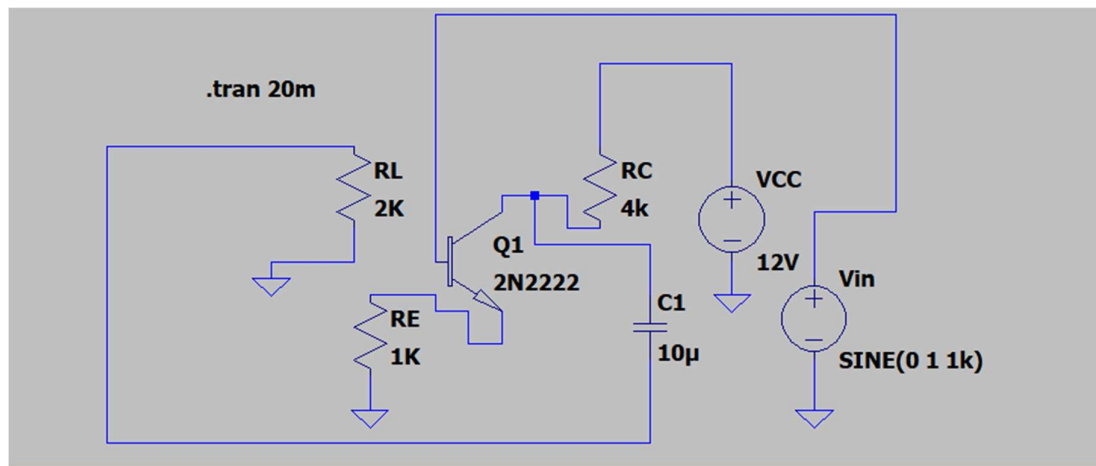
Output coupling capacitor **C2** is mistakenly changed from **10 μF** to **0.01 μF** .

Fault identification and circuit diagram

Smaller capacitor increases the cutoff frequency and attenuates low-frequency signals.



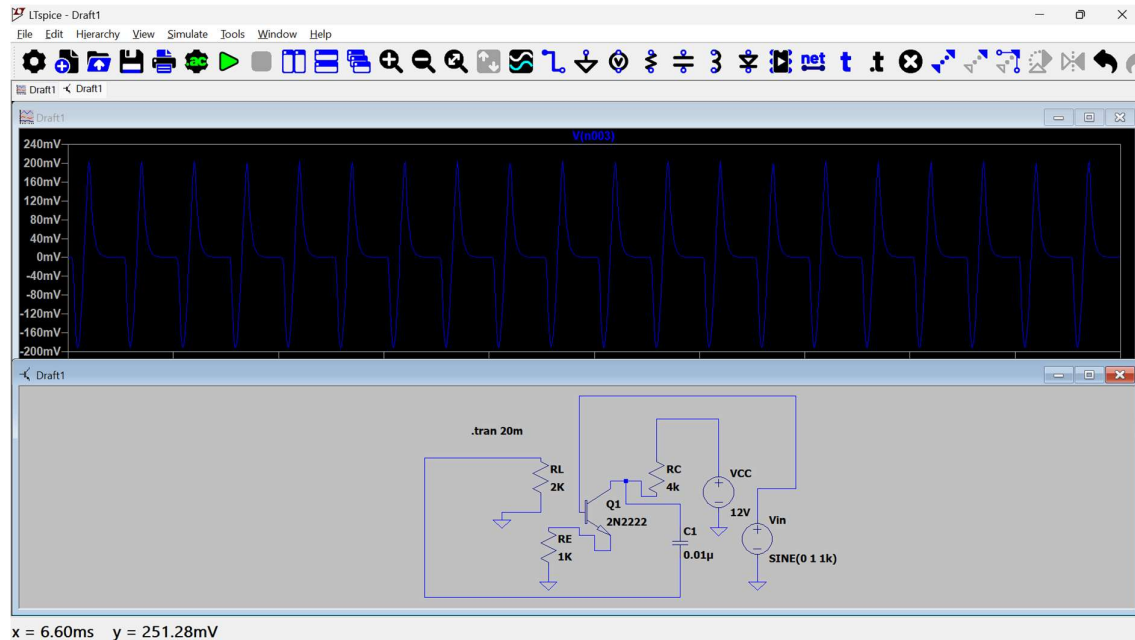
Debugging circuit diagram



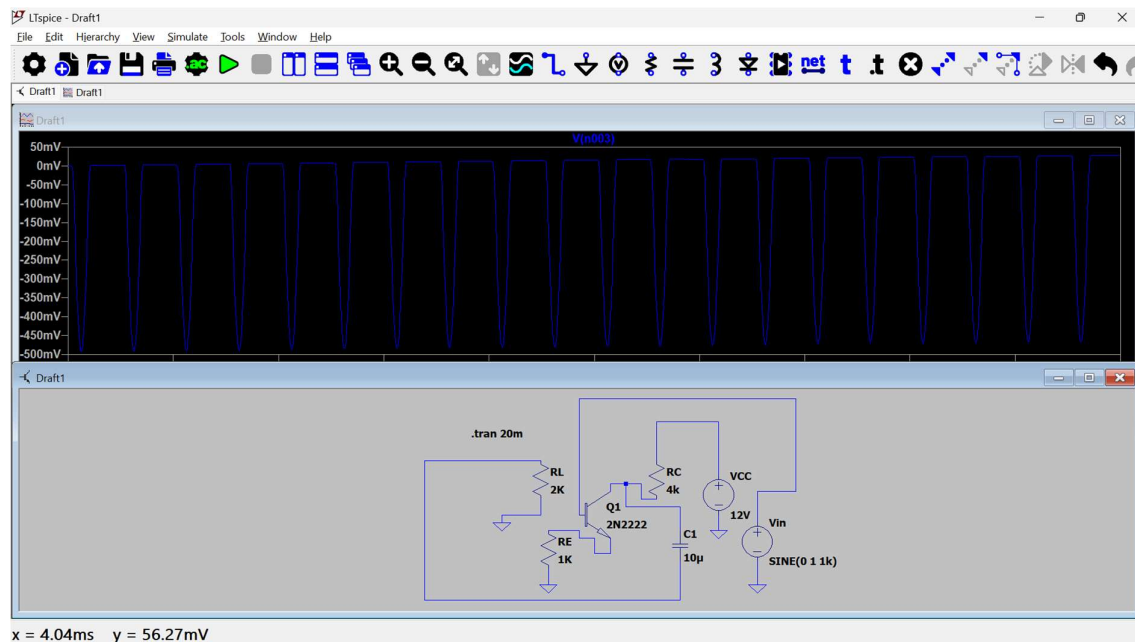
Replace **0.01 μF** with **10 μF**.

Simulation

Faulty output



Correct output



Observation

- ☐ 0.01 μF $\rightarrow$ Low output amplitude, poor low-frequency response.
- ☐ 10 μF $\rightarrow$ Proper output amplitude, good low-frequency response.

Conclusion

The correct coupling capacitor value is **10 μF** because it blocks DC while allowing low-frequency AC signals to pass with minimum attenuation.